A PRELIMENERY REPORT ON

Fake Instagram Profile Identification and Classification Using machine Learning

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
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FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING (COMPUTER ENGINEERING)

SUBMITTED BY

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CERTIFICATE

This is to certify that the project report entitled

"Fake Instagram Profile Identification and Classification Using machine Learning"

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ABSTRACT

Fake Instagram profiles are a growing problem, as they can be used to spread false information, deceive people, and harass and threaten others. The project aims to develop a robust and scalable system for identifying and classifying fake Instagram profiles using machine learning. This system will collect a large and diverse dataset of Instagram profiles, both real and fake. The data will be preprocessed and cleaned, and then relevant features will be extracted. Various machine learning algorithms will be evaluated to select the best model for identifying and classifying fake Instagram profiles. The trained model will be evaluated on a hold-out test set to ensure that it is able to generalize to new data. Once the model is evaluated and deemed satisfactory, it will be deployed in a production environment to identify and classify fake Instagram profiles in real time. The system is expected to have a significant impact on the fight against fake Instagram profiles. By identifying and classifying fake Instagram profiles, the system can help make the social media platform safer and more secure for everyone.

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INTRODUCTION

1.1 Overview

Fake Instagram profiles are a growing problem, as they can be used to spread false information, deceive people, and harass and threaten others. This engineering final year project aims to develop a robust and scalable system to identify and classify fake Instagram profiles using machine learning.The system will collect a large and diverse dataset of Instagram profiles, both real and fake. The data will be preprocessed and cleaned, and then relevant features will be extracted. Various machine learning algorithms will be evaluated to select the best model for identifying and classifying fake Instagram profiles.The trained model will be evaluated on a hold-out test set to ensure that it is able to generalize to new data. Once the model is evaluated and deemed satisfactory, it will be deployed in a production environment to identify and classify fake Instagram profiles in real time.The system is expected to have a significant impact in the fight against fake Instagram profiles. By identifying and classifying fake Instagram profiles, the system can help make the social media platform safer and more secure for everyone.

Here are some of the main advantages of the system:

• Better safety and security for social media users

• Less spread of misinformation and misinformation

• Reduced risk of online scams and fraud

• Improve user experience by reducing the visibility of fake profiles

The system is expected to be used by various stakeholders including social media platforms, law enforcement agencies and cyber security companies.

1.2 Motivation

Fake Instagram profiles have become a huge problem, posing a threat to the safety and security of the social media platform. Machine learning offers a promising approach to overcome this challenge by identifying and classifying fake profiles based on specific patterns and characteristics.This engineering final year project aims to develop a robust and scalable machine learning model to accurately identify and classify fake Instagram profiles in real time.

The project will involve collecting and preprocessing a large and diverse dataset of Instagram profiles, both real and fake. The dataset will then be used to train various machine learning algorithm for identifying and classifying fake Instagram profiles in real time will be selected and deployed in a production environment.This project is motivated by the desire to make social media platforms safer and more trustworthy for everyone. By identifying and classifying fake Instagram profiles, this project can help reduce the spread of misinformation, scams and harassment. This will create a more positive and enjoyable experience for all users.

1.3 Problem Definition and Objectives

- To achieve an accuracy of at least 95% in identifying and classifying fake Instagram profiles.
- To reduce the false positive rate to less than 1%.
- To develop a system that can be deployed in a production environment and scaled to handle millions of Instagram profiles.
- To identify different types of fake Instagram profiles, such as bot accounts, impostor accounts, and spam accounts.
- To develop techniques to detect new and emerging types of fake Instagram profiles.
- To improve the performance of fake profile identification models by using new and innovative machine learning techniques.
- To develop a robust and scalable machine learning model that can accurately identify and classify fake Instagram profiles in real time.
- To improve the state-of-the-art in fake Instagram profile identification and classification using machine learning.

1.4 Project Scope & Limitations

Scope:

- Better safety and security for social media users
- Less spread of misinformation and misinformation
- Reduced risk of online scams and fraud
- Improve user experience

Limitations:

- Variability in Profile Characteristics
- Labelled Dataset
- Data Privacy Concern

CHAPTER 2

LITERATURE REVIEW

2.1Literature Survey:

- Towards conducting responsible research with teens and parents regarding online risks by Zainab Agha, Neeraj Chatlani, Afsaneh Razi, and Pamela Wisniewski (2020).This paper discusses the importance of conducting responsible research with teens and parents regarding online risks. The authors argue that it is important to involve teens and parents in the research process, and to take steps to protect their privacy and safety
- Understanding the Digital Lives of Youth: Analyzing Media Shared within Safe Versus Unsafe Private Conversations on Instagram by Shiza Ali, Afsaneh Razi, Seunghyun Kim, Ashwaq Alsoubai, Joshua Gracie, Munmun De Choudhury, Pamela J Wisniewski, and Gianluca Stringhini (2022).This paper presents a mixed-method analysis of the media files shared within safe and unsafe private conversations on Instagram. The authors found that significantly more media files were shared in safe conversations than in unsafe conversations.
- Detect fake profiles on social media network by Telangana Today (2023): This article discusses the different ways in which fake profiles can be detected on social media networks. The author mentions that fake profiles can be detected by analyzing the profile picture, the number of followers and followings, the content posted by the user, and the user's behavior on the platform. The author also discusses the challenges of detecting fake profiles, and the need for developing more sophisticated methods for detection.
- Social media? Get serious! Understanding the functional building blocks of social media by Jan H. Kietzmann, Kristopher Hermkens, Ian P. McCarthy, and Bruno S. Silvestre (2011): This paper provides a framework for understanding the functional building blocks of social media. The authors identify six key building blocks: identity, conversations, sharing, groups, presence, and relationships. The authors argue that these building blocks can be used to understand the different types of social media platforms, and to develop new and innovative social media applications.
- Detection of Fake Profiles in Social Media by Aleksei Romanov, Alexander Semenov, Oleksiy Mazhelis, and Jari Veijalainen (2017):

This paper presents a survey of the different methods that have been proposed for detecting fake profiles in social media. The authors classify the methods into three categories: feature-based methods, graph-based methods, and machine learning-based methods. The authors also discuss the challenges of detecting fake profiles, and the need for developing more robust and effective methods. Overall, the literature review suggests that there is a growing interest in the problem of fake profiles on social media. Researchers are developing a variety of methods for detecting fake profiles, but there are still a number of challenges that need to be addressed. One of the key challenges is the development of methods that are robust to new and emerging types of fake profiles.

2.2 Literature Review

Sr. No.	Topic Name	Year	Author Name	Advantages	Disadvantages	Description
1	Understanding the Digital Lives of Youth: Analyzing Media Shared within Safe Versus Unsafe Private Conversations on Instagram	2023	Shiza Ali, Afsaneh Razi	Researchers and policymakers can use this information to develop targeted interventions	Findings from this type of analysis may not be representative of all youth on Instagram.	We performed a mixed-method analysis of the media files shared privately in these conversations to gain human-centered insights into the risky interactions experienced by youth.
2	Instagram Data Donation: A Case Study on Collecting Ecologically Valid Social Media Data for the Purpose of Adolescent Online Risk Detection	2022	Afsaneh Razi, Ashwaq AlSoubai	Over time, this data can be used to track changes in online behaviors, helping to identify trends.	The evolving landscape of social media platform policies and regulations, which may limit data access and usage.	The study aims to create a comprehensive understanding of the online behaviors, interactions, and content consumption patterns of adolescents on the platform.

Sr. No.	Topic Name	Year	Author Name	Advantages	Disdvantages	Description
3	Sliding into My DMs: Detecting Uncomfortable or Unsafe Sexual Risk Experiences within Instagram Direct Messages Grounded in the Perspective of Youth	2023	AFSANEH RAZI, ASHWAQ ALSOUBAI	Platforms like Instagram, understanding how youth engage in conversations .	Research involving the analysis of private messages on a social media platform may raise ethical concerns related to privacy, consent.	Risk detection and adolescent online safety literature through our human-centered approach of collecting and ground truth coding private social media conversations of youth for the purpose of risk classification.
4	Happiness and Sadness in Adolescents' Instagram Direct Messaging: A Neural Topic Modelling Approach	2022	Tim Verbeij* , Ine Beyens, Damian Trilling	Neural topic modeling can process large volumes of text data quickly and efficiently.	Analyzing private messages on social media platforms raises ethical concerns regarding privacy and consent.	There are temporal trends in the expression of happiness and sadness in adolescents' DMs and there are individual differences in these trends.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1.1 Assumptions and Dependencies

Assumptions:

- The users of the software have a basic understanding of Instagram and fake profile detection concepts.
- The system will be deployed on a platform that supports Python and the required libraries for machine learning and GUI development.

Dependencies:

- Python 3.x environment
- Required libraries: Tkinter for GUI development, scikit-learn for machine learning, numpy, pandas, etc.
- Internet connectivity for accessing Instagram profiles (optional).

3.1.2 Functional Requirements

User Interface:
The software shall provide a user-friendly interface developed using Tkinter for ease of interaction.
It shall include options for inputting Instagram profile data such as profile picture, bio information, and posting history.
.

Data Processing:
Upon receiving profile data, the software shall preprocess and extract relevant features using machine learning techniques.
It shall handle missing or incomplete data gracefully and provide appropriate error messages.

Fake Profile Detection:
The system shall employ the Random Forest algorithm for classification of profiles into fake and genuine categories.
It shall provide real-time feedback on the likelihood of a profile being fake based on the trained model.

Result Presentation:
The software shall present the detection results in a clear and understandable format, indicating the probability or confidence level of the prediction.
It shall provide visual indicators or alerts to highlight suspicious profiles .

3.1.3 Nonfunctional Requirements

Accuracy:
The system shall strive to achieve high accuracy in fake profile detection, minimizing false positives and false negatives to maintain user trust.

Reliability:
The software shall be robust and reliable, capable of handling unexpected errors or failures gracefully without compromising the detection process.

Performance:
The software shall be able to process profile data and provide detection results within a reasonable time frame, ensuring a responsive user experience.

Security:
The system shall implement appropriate security measures to protect user data and ensure confidentiality during profile analysis and detection.

Usability:
The user interface shall be intuitive and easy to navigate, catering to users with varying levels of technical expertise.

3.2 System Requirements

3.2.1 Database Requirements

MySql Lite: The system utilizes MySql Lite as the backend database for storing andmanaging login detail of users.

3.2.2 Software Requirements

- Python
- Spyder IDE
- Windows 10
- Random Forest Classifier

3.2.3 Hardware Requirements

- Processor : Core I3
- Ram : 2Gb
- SSD : 40Mb
- Monitor : LCD/LED
- Mouse : Two or Three Button Mouse
- Key Board : Standard Windows Keyboard

3.2.4 Analysis Models : SDLC Model to be applied

For the "Fake Instagram Profile Identification and Classification using Machine Learning " project, the Agile Software Development Life Cycle (SDLC) model offers a structured yet flexible approach to managing the development process. Agile methodologies emphasize iterative development, collaboration, and responsiveness to change, making it well-suited for projects with evolving requirements and complex functionalities.

In this project, the Agile approach involves breaking down the project into smaller iterations or sprints, typically lasting 2-4 weeks.

Each sprint focuses on implementing and testing specific features or functionalities related to fake Instagram Profile.

User stories are used to capture the functional and non-functional requirements from the perspective of end-users, with a prioritized product backlog guiding development efforts. Continuous integration practices ensure that code changes are integrated frequently and tested automatically, with regular regression testing to maintain the integrity of the system.

Incremental releases are delivered at the end of each sprint, incorporating new featuresand improvements based on stakeholder feedback and prioritization. Regular demonstrations and reviews gather feedback from stakeholders and end-users, fostering collaboration and transparency throughout the development process.

Adaptability and flexibility are key principles of Agile, allowing the project team to embrace change and respond to evolving requirements, technology advancements, and user feedback. Robust quality assurance processes are in place to ensure the reliability,accuracy, and security of the system, with risk management strategies addressing potential uncertainties and challenges.

CHAPTER 4

SYSTEM DESIGN

4.1System Architecture

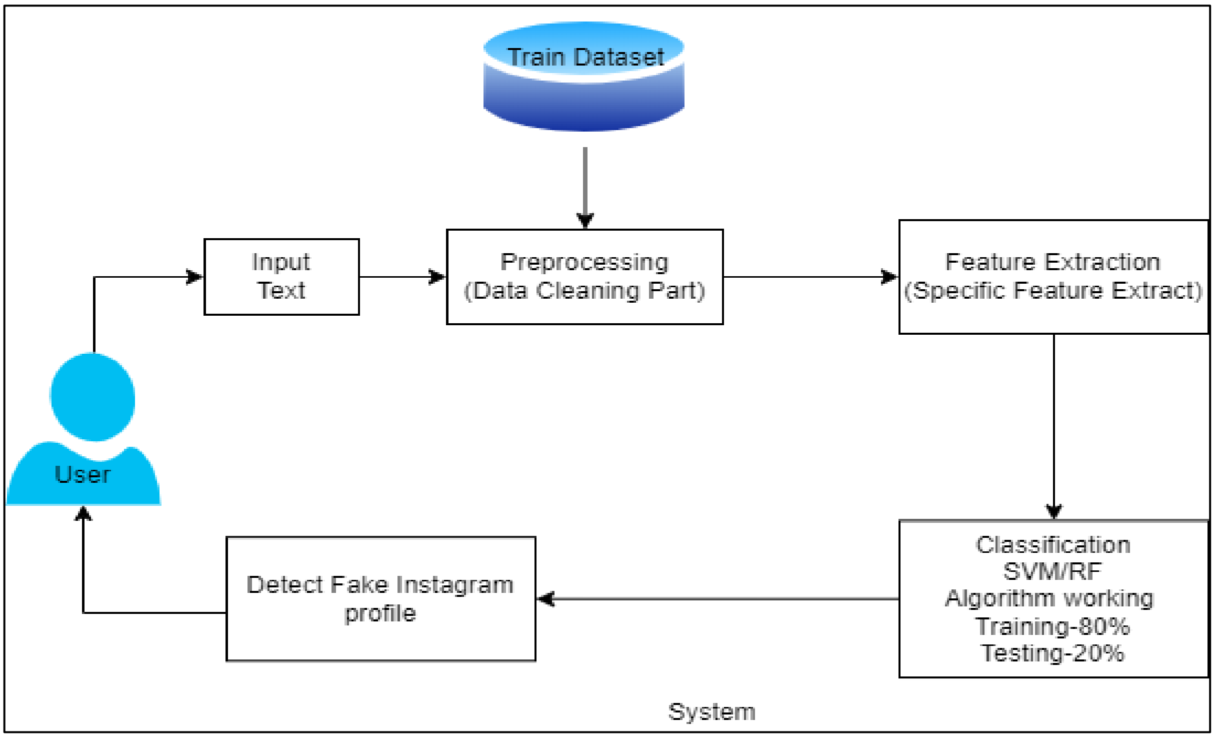
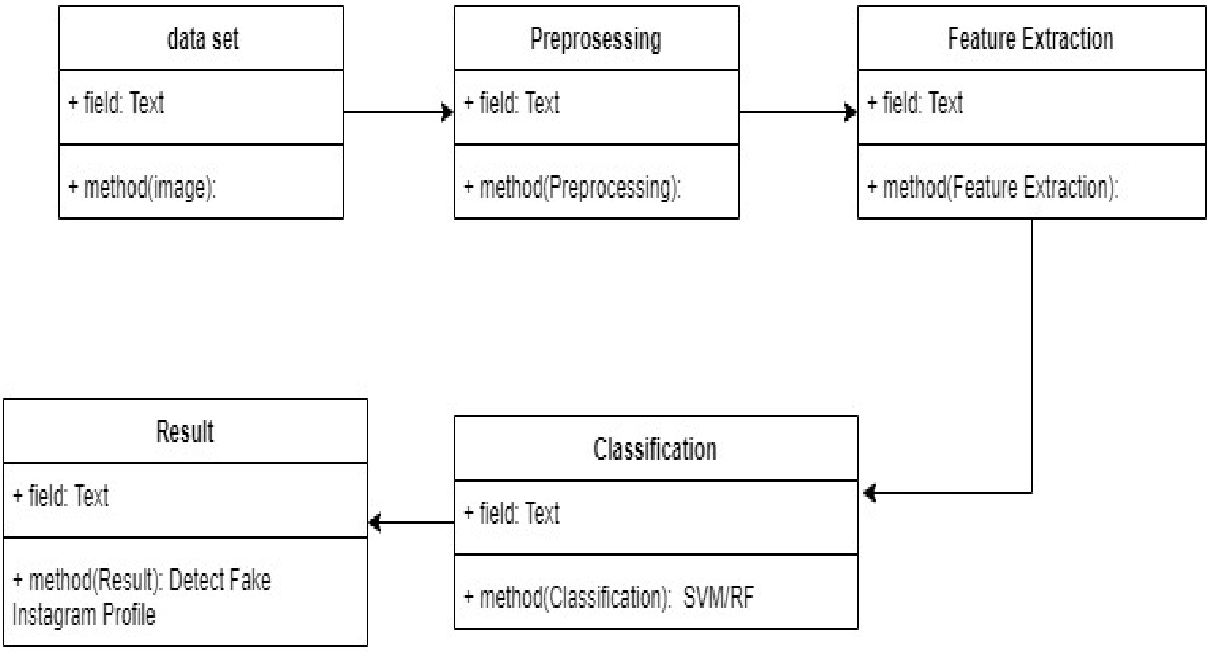


Fig 4.1 System Architecture



4.2 Mathematical Model

- Set Let S be the Whole system S= I,P,O
- I-input
- P-procedure
- O-output
- Input(I)
- I=Text
- Where,
- Dataset- Image dataset Using SVM/RF Algorithm
- Procedure (P),
- P=I, Using I System Detect Fake Instagram profile

In the mathematical model provided:

- The entire system is represented as S, comprising three fundamental elements: input (I),
  procedure (P), and output (O). Thus, S = I, P, O.

- I signifies the input to the system.In this context, the input(I) is specifically defined as "Text".

- The dataset employed within the system pertains to image data. Two distinct algorithms are mentioned for processing this dataset: Support Vector Machine (SVM) and Random Forest
  (RF).

- P denotes the procedure followed within the system. According to the model, the procedure
  - (P) involves utilizing the input (I) to detect fake Instagram profiles. Therefore, the procedure
  - can be articulated as P = I, wherein the system leverages the provided input to identify fraudulent Instagram profiles.

- Within the procedure, the input (I) comprises textual data. This textual information might encompass various attributes associated with Instagram profiles, such as user descriptions, comments, or captions.

- The primary objective of the system is to discern fake Instagram profiles by processing the supplied textual input. This process likely involves employing techniques such as natural language processing (NLP) and machine learning algorithms like SVM or RF to analyze the textual data and make informed determinations regarding the authenticity of the profiles.

4.3 Data Flow Diagrams

In Data Flow Diagram, we Show that flow of data in our system in DFD0 we show
that base DFD in which rectangle present input as well as output and circle show oursystem, In DFD1 we show actual input and actual output of system input of our system is text or image and output is rumour detected likewise in DFD 2 we presentoperation of user as well as admin.

Figure No.4.3.1 Data Flow Diagram-0

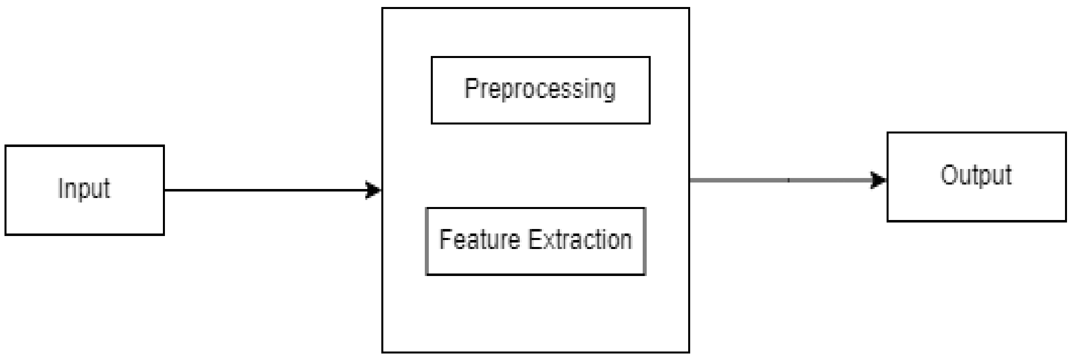


Figure No.4.3.1 Data Flow Diagram-1

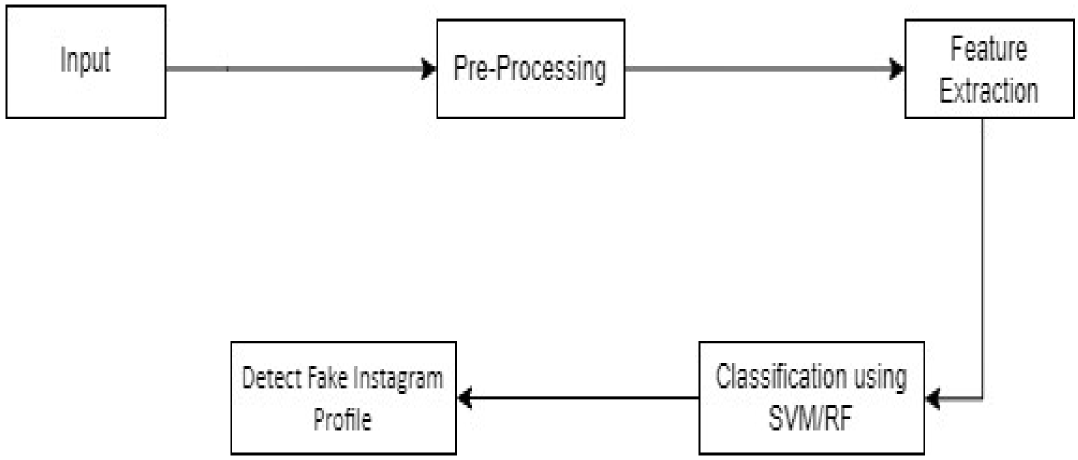


Figure No.4.3.2 Data Flow Diagram-2

4.4 UML Diagrams

Unified Modelling Language is a standard language for writing software blueprints.The UML may be used to visualize, specify, construct and document the artifacts of a software intensive system. UML is process independent, although optimally it should be used in process that is use case driven, architecture-centric, iterative and incremental. The Number of UML Diagrams available.

Use-Case Diagram:

This diagram helps to visualize the interactions between actors and the system's functionalities.

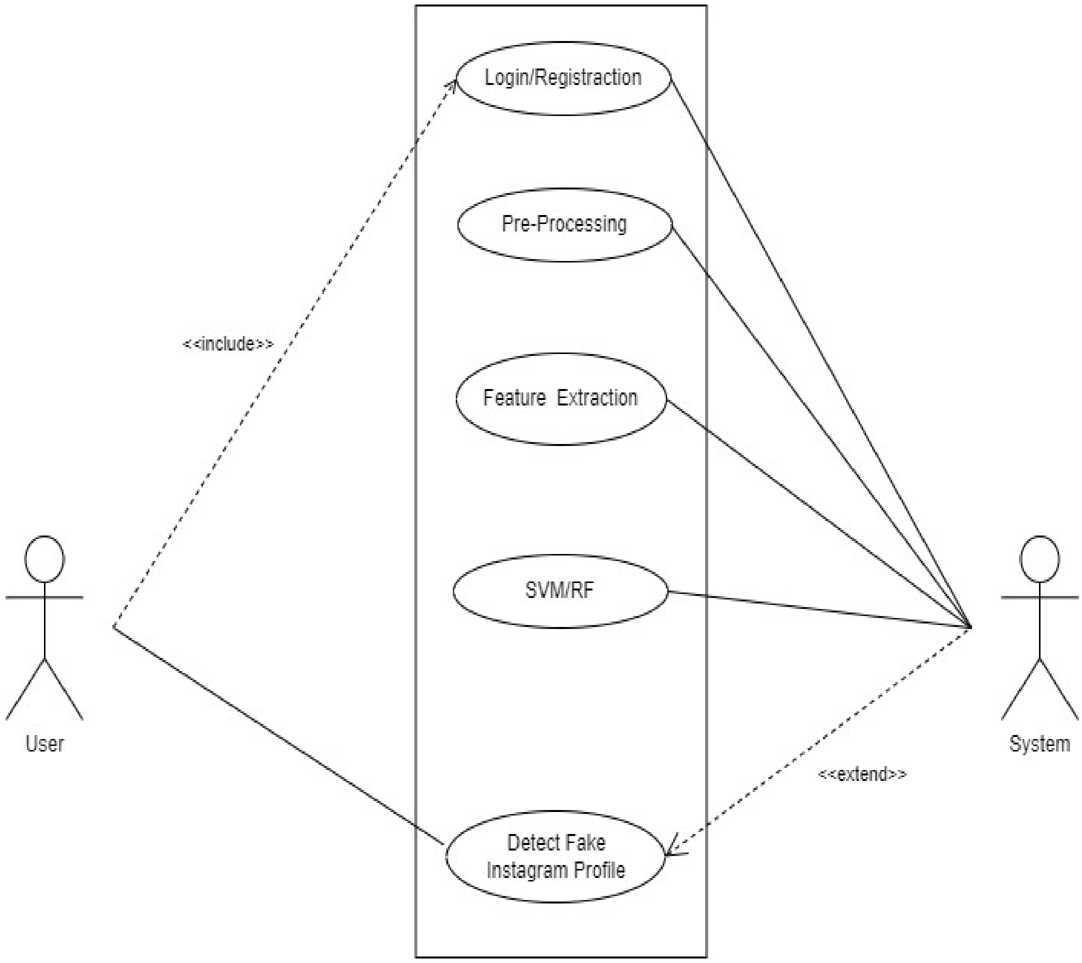


Figure No.4.4.1 Use-Case Diagram

Class Diagram :

A class diagram represents the structure of your system, including classes, their attributes, and methods, as well as how they interact with each other.

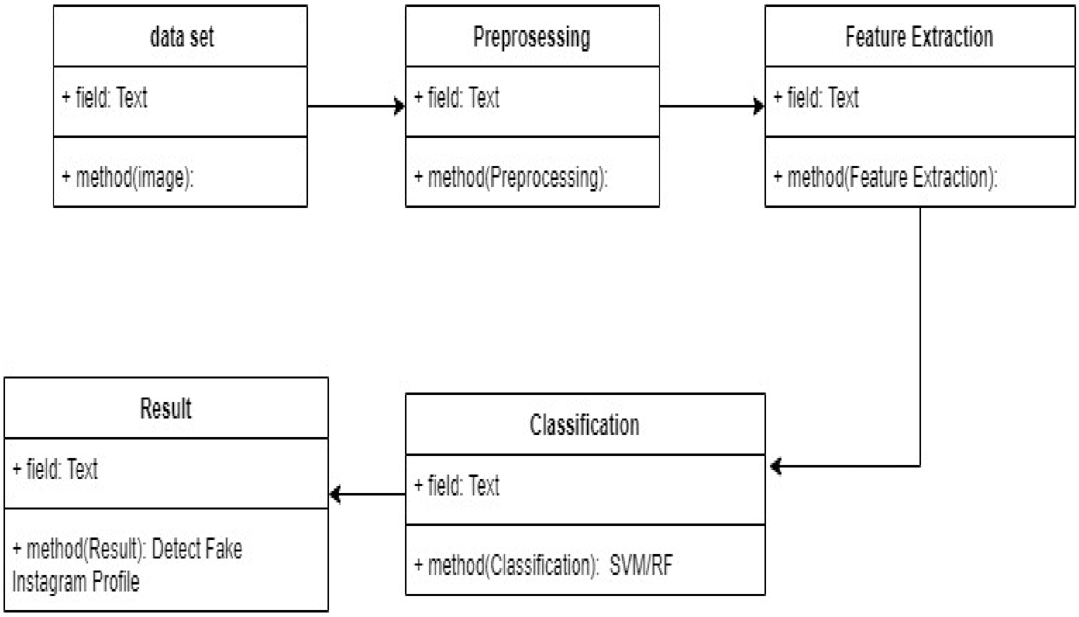


Figure No.4.4.2 Class Diagram

Sequence Diagram:

Here's a basic sequence diagram illustrating the process of a user searching for thenearest shop and viewing offer details.

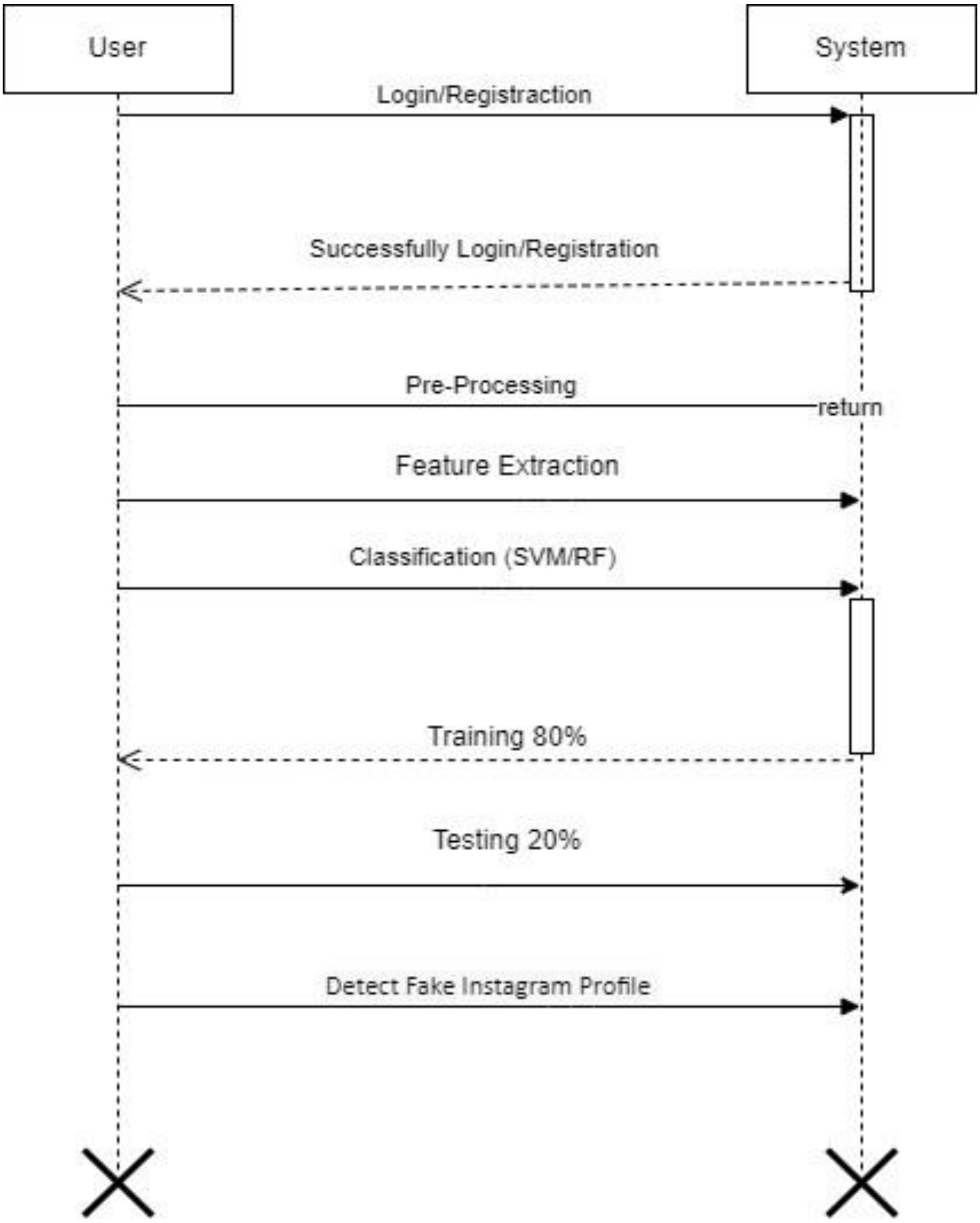


Figure No.4.4.3 Sequence Diagram

Activity Diagram :

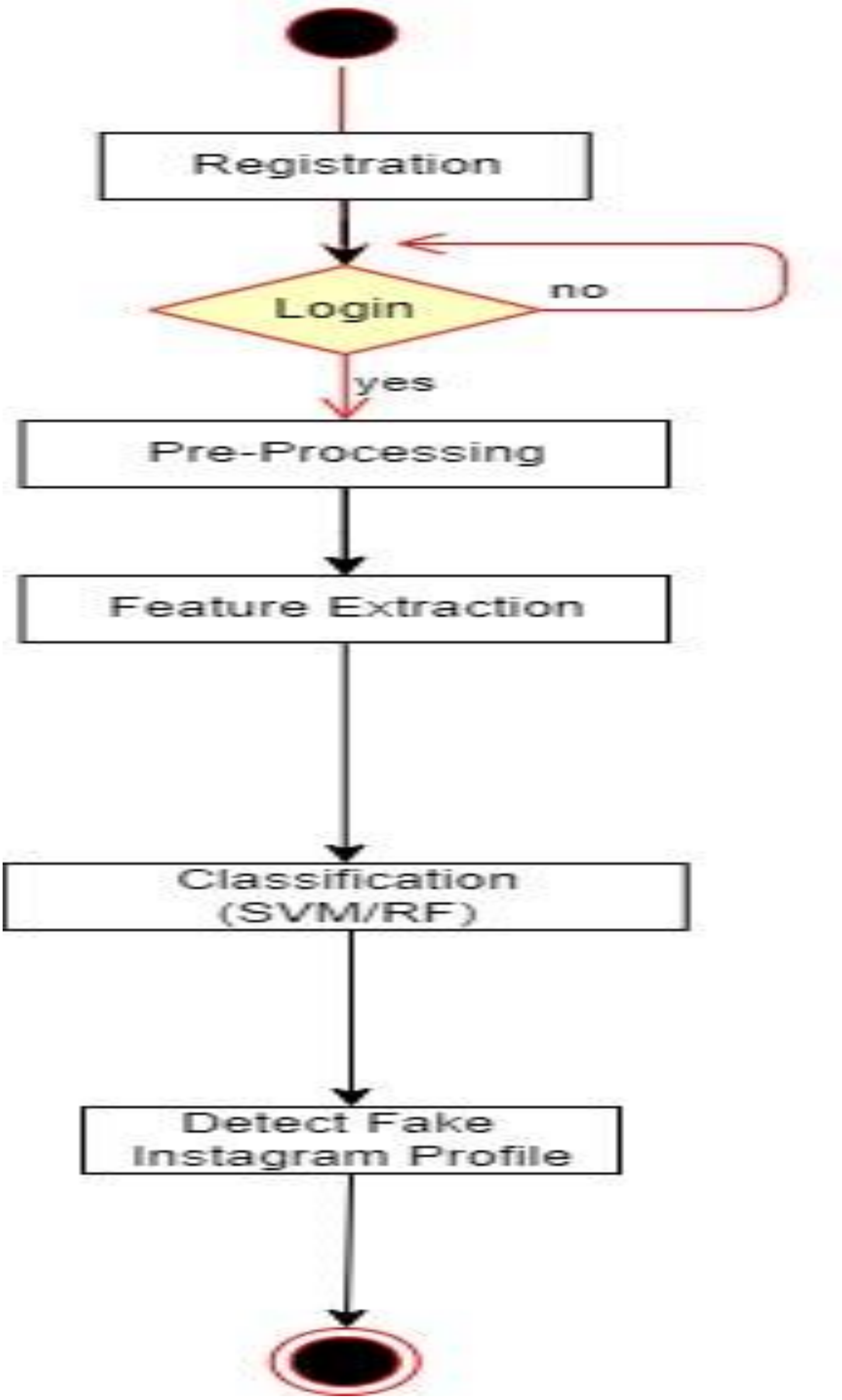


Figure No.4.4.4 Activity Diagram

CHAPTER 5
PROJECT PLAN

5.1 Project Estimate

Work Breakdown Structure (WBS):

Divide the project into smaller tasks or work packages, such as:

- Acquiring and preprocessing the dataset.
- Developing the fake profile detection algorithm.
- Implementing the graphical user interface (GUI) using Tkinter.
- Training and evaluating the machine learning model.
- Conducting thorough testing and validation.
- Documenting the project and preparing reports.

Estimation Techniques:

Utilize various estimation techniques including expert judgment and analysis of past projects to estimate the effort, time, and resources required for each task.

5.2 Risk Management

5.2.1 Risk Identification

Engage in brainstorming sessions and workshops to identify potential risks related to the project.

Risks may include data quality issues, model overfitting, and resource constraints.

5.2.2 Risk Analysis

Analyze identified risks to understand their probability and potential impact on project goals.

Prioritize risks based on severity to focus mitigation efforts effectively.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Collaboratively develop risk mitigation strategies to address identified risks.

Establish regular monitoring mechanisms such as progress meetings to track risks and implement necessary actions to mitigate them.

5.3 Project Schedule

5.3.1 Project Task Set

Identify project tasks based on the project scope, including:

- Data acquisition and preprocessing.
- Algorithm development and model training.
- GUI implementation.
- Testing and validation procedures.
- Documentation and presentation preparation.

5.3.2 Task Network

Visualize task dependencies and relationships to understand the flow of work.

Conduct critical path analysis to identify key tasks and determine the project's minimum duration.

5.3.3 Timeline Chart

July 2023:

- Project Initiation.

- Define project objectives, scope, and deliverables.

- Formulate the project team and assign roles and responsibilities.

August 2023:

- Data Collection and Preprocessing.

- Gather labeled dataset of fake and genuine Instagram profiles.

- Preprocess the dataset to clean and prepare it for training.

September 2023:

- Algorithm Development.

- Research and implement machine learning algorithms for fake profile detection.

- Experiment with feature engineering techniques to enhance model performance.

October 2023:

- GUI Development.

- Design and develop a user-friendly graphical interface using Tkinter.

- Integrate the GUI with the backend algorithms for seamless interaction.

November 2023:

- Testing and Validation.

- Conduct thorough testing of the entire system, including both the algorithm and GUI.

- Validate the system's performance against a separate test dataset to ensure reliability.

December 2023:

- Documentation
- Document the project process, including data preprocessing steps, algorithm implementation, and GUI design.
- Prepare user manuals and guides for future reference.

-

January 2024:

- Presentation Preparation.
- Create a comprehensive presentation summarizing the project's objectives, methodology, and results.
- Rehearse the presentation to ensure clarity and effectiveness

February 2024:

- Presentation and Demonstration
- Deliver the project presentation to stakeholders, including instructors and peers.
- Demonstrate the functionality of the fake Instagram profile detection system.

March 2024:

- User Acceptance Testing (UAT)
- Tested the application on Random Instagram Profiles.


TIMELINE CHART
JULY
·Project Initiation.
-Define project objectives, scope, and deliverables.
Formulate the project team and assign roles and responsibilities
AUGUST
·Data Collection and Preprocessing.
-Gather labeled dataset of fake and genuine Instagram profiles.
-Preprocess the dataset to clean and prepare it for training.
SEPTEMBER
· Algorithm Development.
-Research and implement machine learning algorithms for fake profile detection.
-Experiment with feature engineering techniques to enhance model performance.
OCTOBER
· GUI Development.
-Design and develop a user-friendly graphical interface using Tkinter.
-Integrate the GUI with the backend algorithms for seamless interaction.
NOVEMBER
·Testing and Validation.
-Conduct thorough testing of the entire system, including both the algorithm and GUI.
DECEMBER
· Documentation
-Document the project process, including data preprocessing steps, algorithm implementation, and GUI design.
JANUARY
· Presentation Preparation.
-Create a comprehensive presentation summarizing the project's objectives, methodology,
and results
FEBRURAY
· Presentation and Demonstration
- Demonstrate the functionality of the fake Instagram profile detection system.
MARCH
· User Acceptance Testing(UAT)
- Tested the application on Random Instagram Profiles.

5.1 Team Organization

5.1.1 Team Structure

- Project Manager and Report Creator: Responsible for overseeing the project, creating reports, presentations, and other documentation required for project management and reporting purposes.

- Project Coordinator: Acts as the liaison between the project team and the project guide. Responsible for communicating project requirements, gathering feedback, and ensuring alignment between project objectives and guide expectations.

- Development Team: Comprised of two developers responsible for building the Desktop application. Collaborate closely to implement features, conduct testing, and ensure the app meets quality standards.

5.1.2 Management reporting and communication

- The project manager and coordinator will communicate regularly with the development team to provide updates, gather requirements, and address any challenges or issues.

- The project coordinator will also maintain communication with the project guide, providing progress reports, seeking feedback, and ensuring project alignment with guide expectations.

CHAPTER 6
PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

Software Implementation

1) Python IDE- TO develop all the Programs and Algorithms

Fig.6.1 Python IDE logo

Python IDE is an integrated development environment for Python, which has been bundled with the default implementation of the language since 1.5.2b1.[3][4] It is packaged as an optional part of the Python packaging with many Linux distributions. It is completely written in Python and the Tkinter GUI toolkit (wrapper functions for Tcl/Tk). IDLE is intended to be a simple IDE and suitable for beginners, especially in an educational environment.To that end, it is cross-platform and avoids feature clutter.

2) Tkinter – GUI toolkit

Fig.6.2 Tkinter logo

Tkinter is utilized in your project for its seamless integration with Python, serving as the standard GUI toolkit for Python applications. Its user-friendly nature and ease of use make it an ideal choice for developing the graphical interface of your fake Instagram profile detection system. With Tkinter, you can create a cross-platform compatible interface that offers customization options to tailor the user experience according to your project's requirements. Additionally, Tkinter's strong community support provides access to resources and assistance, ensuring a smooth development process for your application.

3) Sql Lite

Fig.6.3 SQLite logo

SQLite is chosen as the database management system to securely store login credentials within the application. Its lightweight nature and seamless integration with Python ensure efficient data persistence and confidentiality. This decision aligns with the project's goals of simplicity, reliability, and security, providing a robust foundation for user authentication while maintaining a streamlined development process.

4) Random Forest(RF)

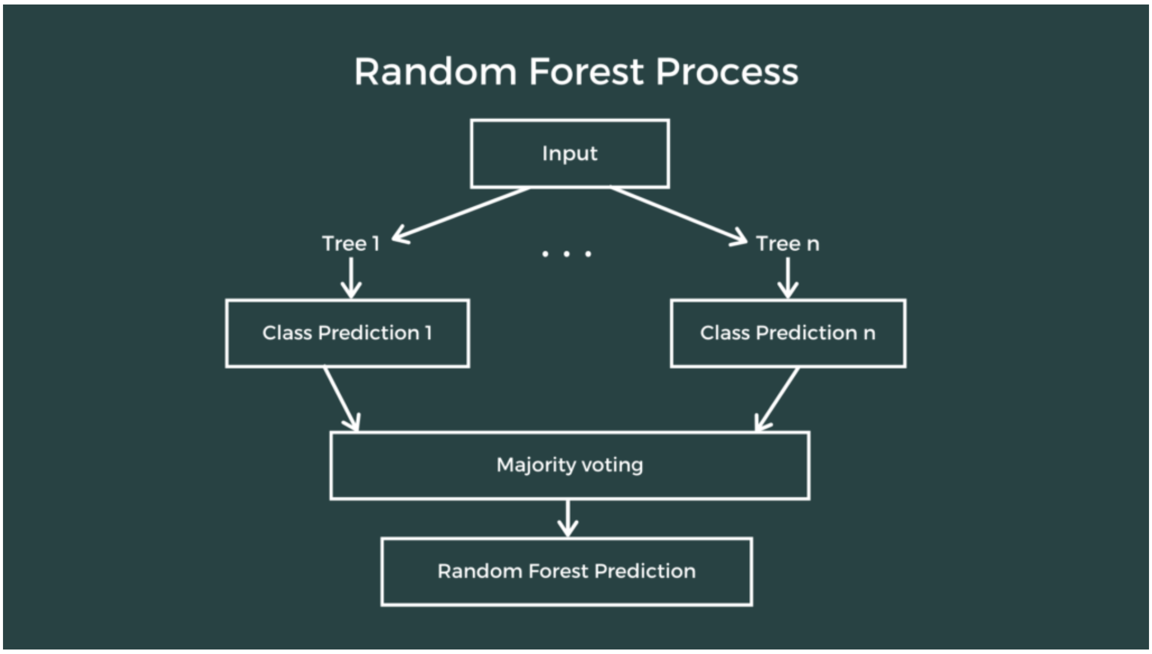


Fig.6.4 Random Forest Algorithhm

Random Forest algorithm enhances the project's ability to detect fake Instagram profiles by aggregating predictions from multiple decision trees, mitigating overfitting, and improving generalization. Its capacity to handle high-dimensional data and identify key features aids in discerning patterns indicative of fake profiles, contributing significantly to the project's objective of developing a reliable detection mechanism.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

- Unit Testing: Individual components such as functions and methods are tested inisolation to ensure they perform as expected.
- Integration Testing: Multiple components are combined and tested together to verifytheir interactions and interfaces.
- System Testing: The entire system is tested as a whole to validate its functionality,performance, and reliability.
- User Acceptance Testing (UAT): End-users evaluate the system to ensure it meetstheir requirements and expectations.

7.2 Testing strategies & Test Procedures

Testing Strategies:

Data Splitting: The data is divided into training and testing sets. The model is trained on the training set (typically 80% of the data) and evaluated on the unseen testing set (20% of the data). This approach helps assess the model's ability to generalize to new data. Techniques like k-fold cross-validation can also be explored for a more robust evaluation.

Error Metrics: Clearly defined metrics are essential to gauge the model's performance. Here are some relevant metrics for classification tasks:

Accuracy: Overall percentage of correctly classified profiles (fake and genuine).

Precision: Ratio of correctly identified fake profiles out of all profiles identified as fake.

Recall: Ratio of correctly identified fake profiles out of all actual fake profiles in the testing data.

F1-score: Harmonic mean between precision and recall, useful for imbalanced datasets.

Baseline Model: A baseline model is established to serve as a benchmark for comparison. This could be a simple model (e.g., predicting all profiles as fake or genuine) or a different machine learning algorithm trained on the same data.

Testing Procedures:

Preprocessing Test Data: The testing data undergoes the same preprocessing steps (cleaning, normalization, etc.) applied to the training data.

Training the Model: The Random Forest model is trained on the designated training set with chosen hyperparameters (number of trees, maximum depth, etc.).

Evaluation on Testing Set: The trained model is used to make predictions on the unseen testing set. The chosen error metrics (accuracy, precision, recall, F1-score) are calculated to understand how well the model generalizes to unseen data.

Error Analysis: Misclassified profiles are analyzed to identify patterns in errors. Look for specific features associated with misclassifications to improve the model or data collection.

Comparison to Baseline: The performance of the Random Forest model is compared to the baseline model. This comparison helps assess the effectiveness of the chosen approach.

CHAPTER 8
RESULTS

8.1 User Interface:

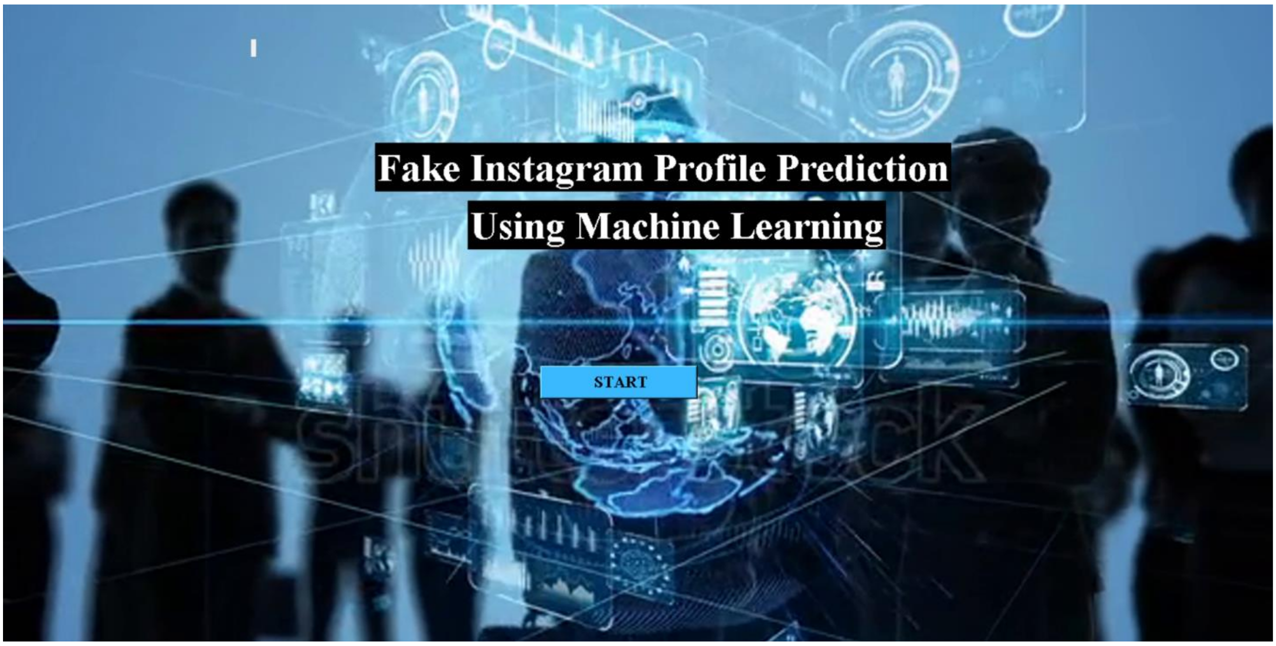


Fig.8.1 Start Up Page

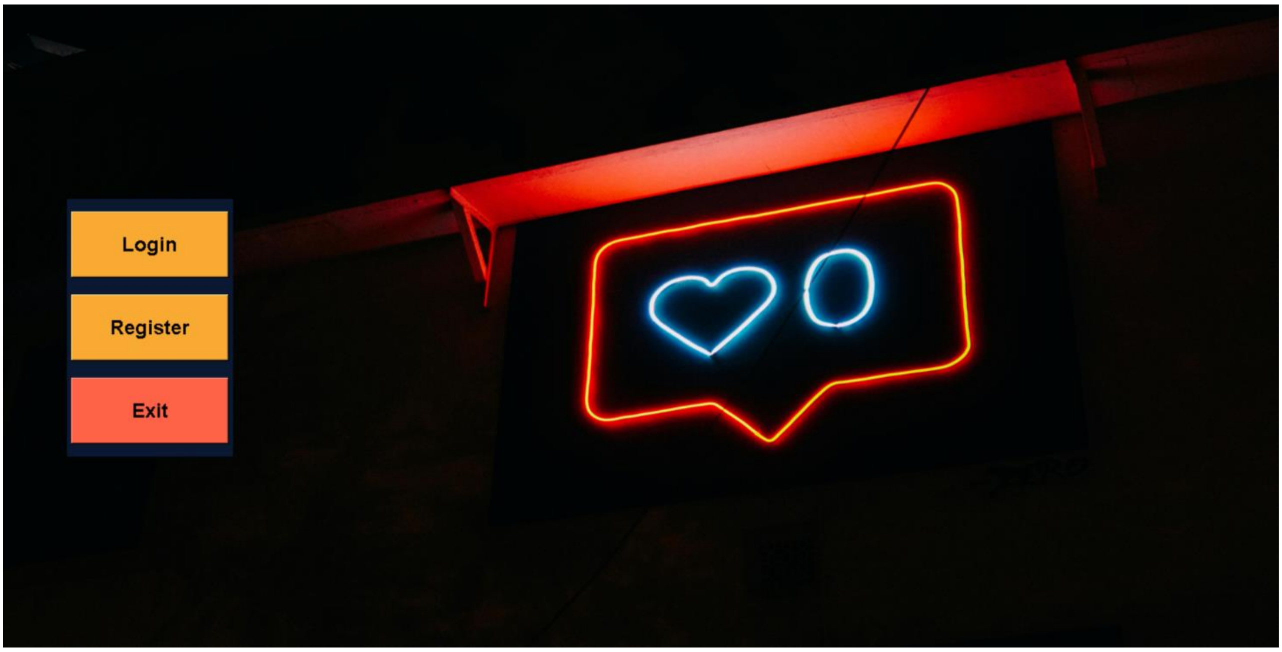


Fig.8.2 Registration Page

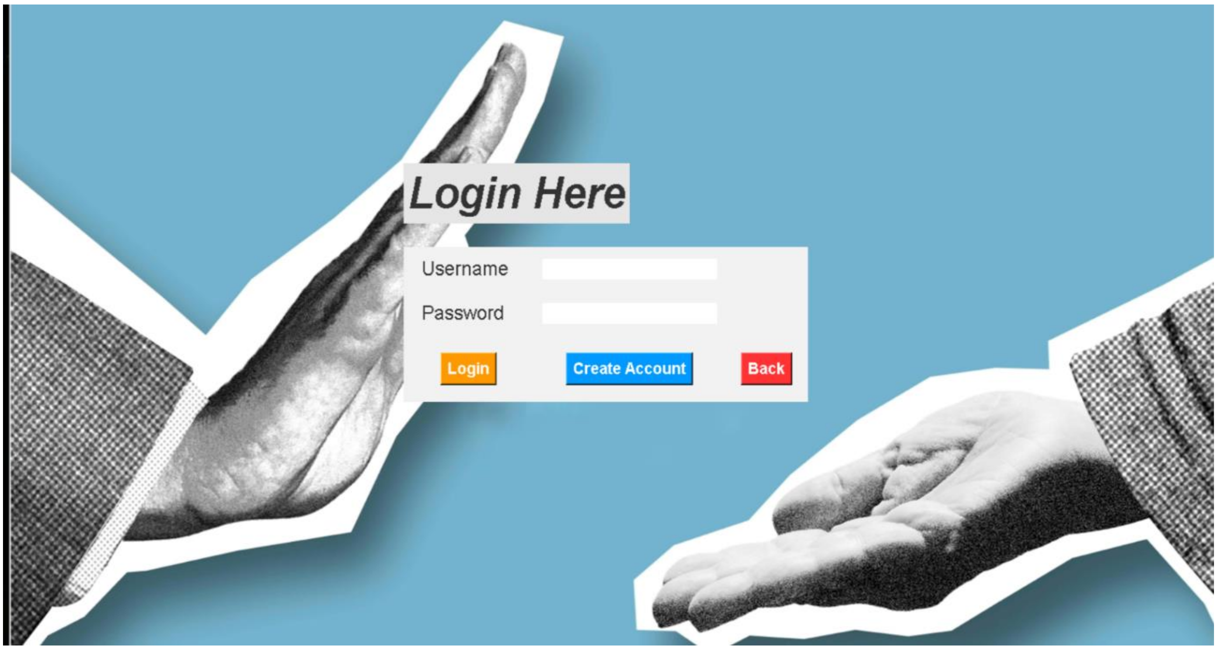


Fig.8.3 Login Page

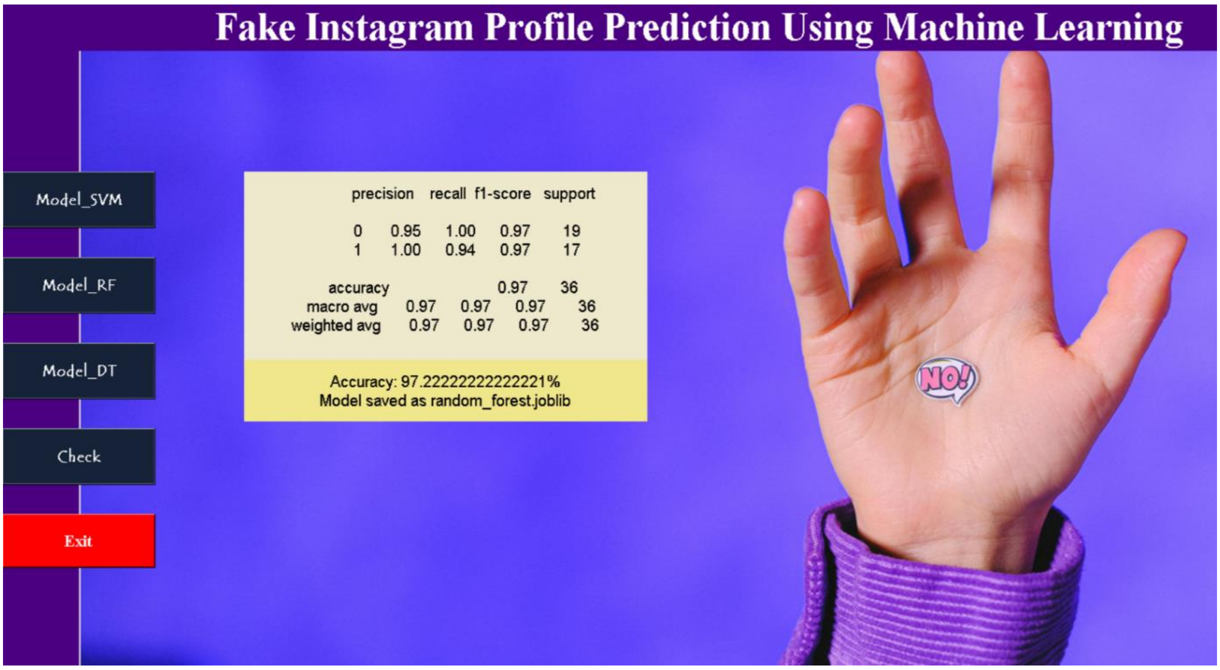


Fig.8.4 Model Selection Page

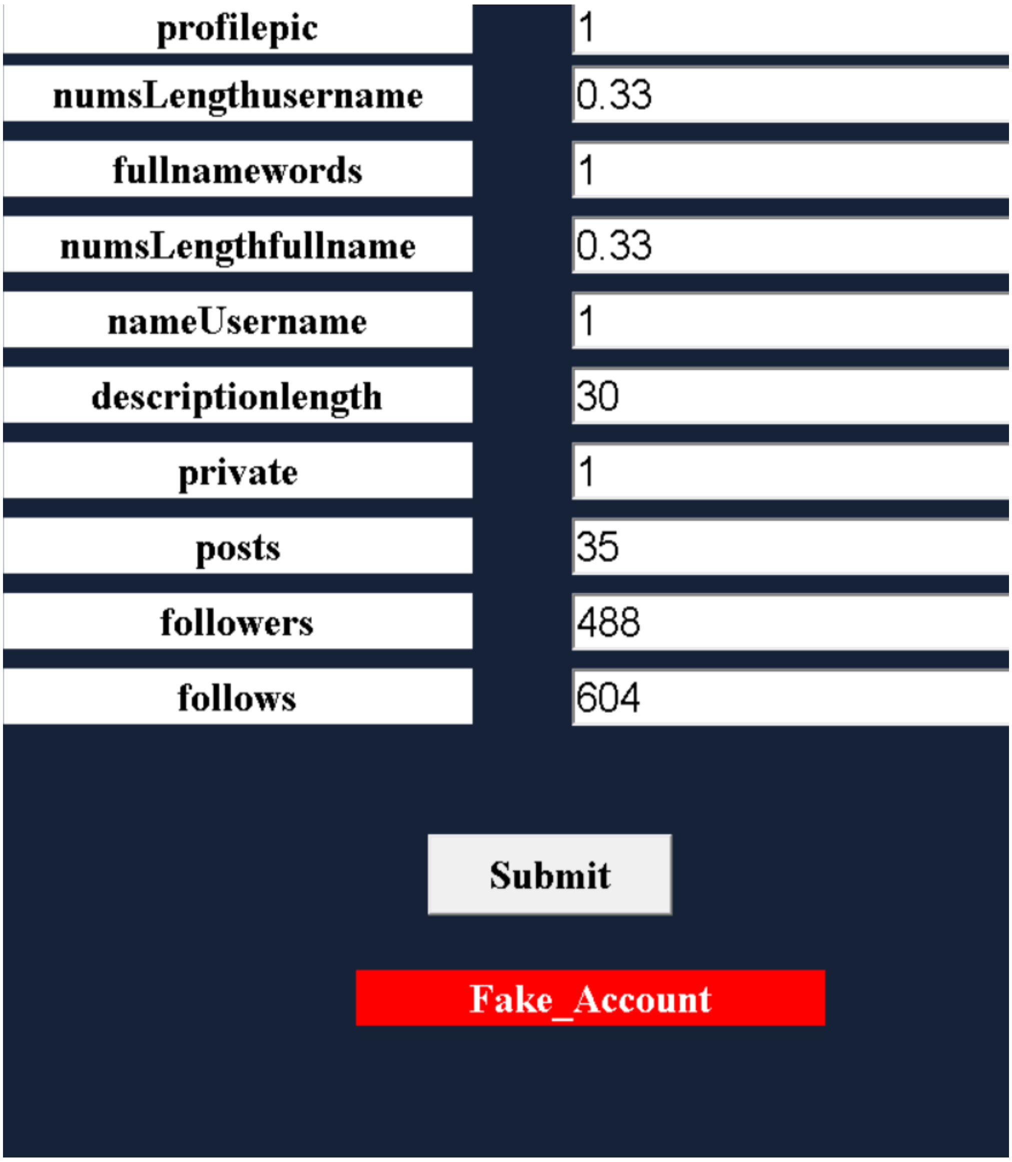


Fig.8.4 Checking Page

In this scenario, the Random Forest model successfully classifies the Instagram profile as
fake. This is indicated by Red Pop up that occurs whenever a fake profile is detected. This demonstrates the model's ability to accurately identify profiles that exhibit characteristics associated with being fake.

By leveraging this model, Instagram can potentially automate the detection of fake profiles, improving platform security and user experience. When a profile is flagged as fake, further investigations or restrictions can be implemented to maintain a genuine user base.

CHAPTER 9
CONCLUSIONS

9.1 Conclusions and Future scope:

Fake Instagram profiles have become a major problem in recent years, as they can be used to spread misinformation, scam people, and harass and bully others. Machine learning has emerged as a promising approach to identifying and classifying fake Instagram profiles.This research presents a comprehensive approach to tackle the persistent issue of fake profiles on social media platforms, with a specific focus on Instagram.

By leveraging the power of machine learning techniques, this research contributes to creating a safer and more trustworthy online environment for users, bolstering user confidence, and upholding the integrity of social media community.The research's outcomes extend beyond the realm of academia, impacting the lives of individuals, businesses, and society as a whole.

One area of future scope is to develop more sophisticated machine learning models that can identify and classify fake Instagram profiles with even greater accuracy. This could be achieved by using larger and more diverse datasets of fake and real Instagram profiles, as well as by exploring new machine learning techniques.Another area of future scope is to develop real-time monitoring systems that can identify and flag fake Instagram profiles as soon as they are created.

This would help to prevent fake profiles from spreading misinformation, scamming people, and harassing and bullying others.Finally, it is important to consider the financial resources required for data collection, machine learning model development, real-time monitoring infrastructure, and system maintenance.

The potential costs associated with the development and maintenance of the system should be evaluated against the benefits of reducing the impact of fake profiles on Instagram.Overall, the research on "Fake Instagram Profile Identification and Classification using Machine Learning" presents a promising approach to tackling the issue of fake profiles on social media platforms.
Future research should focus on developing more sophisticated machine learning models, real-time monitoring systems, and cost-effective solutions to identify and classify fake Instagram profiles. By addressing these challenges, we can help to make social media platforms safer and more enjoyable for everyone.

9.2 Applications

Content Verification: Helps verify the authenticity of information shared on Instagram by suspicious profiles.
Electoral Integrity: Assists in mitigating manipulation attempts by fake profiles during elections.
Trend Analysis: Identifies emerging trends in social media manipulation for proactive countermeasures.
Law Enforcement Support: Potentially assists law enforcement in identifying fake profiles linked to criminal activity.

CHAPTER 10

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